

SYSTEMATIC REVIEW

Open Access



Efficacy of daily versus alternate day oral iron supplementation for management of anaemia among general population: a systematic review and meta-analysis

Anupkumar D. Dhanvijay¹ , Vipin Patidar⁴ , Jayvardhan Singh² , Subodh Kumar³ , Shiv Kumar Mudgal^{4*} , Seshadri Reddy Varikasuvu⁵ and Rajesh Kumar⁶

Abstract

Background Iron deficiency anemia (IDA) remains a prevalent global health issue. While oral iron therapy is the first-line treatment, the optimal dosing strategy—daily versus alternate-day—remains debated, especially for general population use. Therefore, this review aimed to compare the efficacy and tolerability of daily versus alternate-day oral iron supplementation for IDA in the general population.

Methods Searches were conducted across major databases through March 2025. Risk of bias was evaluated utilizing the Cochrane RoB 2 tool, and the certainty of evidence was graded using GRADE. The primary outcome was change in hemoglobin concentration; secondary outcomes included serum iron, ferritin, transferrin saturation, TIBC, MCV, and adverse events. R Studio software, version 4.2.3, and RevMan used for all analyses.

Results This SRMA (systematic review and meta-analysis) included 11 RCTs involving 1014 participants. The pooled analysis found a small, statistically non-significant increase in hemoglobin with daily dosing over alternate-day (MD: 0.28, 95% CI: -0.01 to 0.56, $p=0.06$, $z=1.91$). Secondary outcomes revealed no significant differences among groups for serum iron, ferritin, transferrin saturation, TIBC, and MCV. Adverse effects were similar between groups (RR: 1.07, 95% CI: 0.86 to 1.34), though metallic taste was more frequent with daily dosing. Risk of bias was low to moderate across studies. Certainty of evidence was rated very low for most outcomes due to heterogeneity and imprecision.

Conclusion Both daily and alternate-day oral iron supplementation are comparably effective for treating IDA, with alternate-day dosing showing better tolerability. Due to low certainty in evidence, treatment decisions should be individualized pending further high-quality research.

Clinical trial number Not applicable.

Keywords Iron deficiency anemia, Oral iron supplementation, Daily dosing, Alternate-day dosing, Meta-analysis

*Correspondence:
Shiv Kumar Mudgal
shiv.nur@aiimsdeoghar.edu.in

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

Anemia affects about one-third of the global population and contributes to 8.8% of all health-related disabilities worldwide [1]. In 2019, approximately 40% of children aged 6–59 months, 30% of non-pregnant women, and 36% of pregnant women aged 15–49 years were anaemic [2]. The number of anemia cases grew from 1.5 billion in 1990 to 1.92 billion in 2021, making it the leading cause of years lived with disability (YLDs) that year [3]. Dietary iron deficiency remains one of the main causes [4], with iron-deficiency anemia alone responsible for 22% of maternal deaths and 50 million years of healthy life lost in 2019 [5].

According to the World Health Organization (WHO), anemia is defined as haemoglobin levels below 13 g/dL in men and 12 g/dL in women [6]. Iron deficiency anemia (IDA) — the most common type — typically shows up as hypochromic microcytic anemia and is diagnosed by low serum ferritin and iron levels, increased total iron-binding capacity, and low transferrin saturation [7]. Adults normally lose about 1–2 mg of iron per day — a small fraction of total body stores — and this needs to be replenished through diet. Daily iron requirements are 8 mg for adult men and 18 mg for menstruating women [8].

The main goal in treating IDA is to correct haemoglobin levels, restore red cell indices, and rebuild iron stores. Management should be individualized, with oral iron supplements typically being the first choice for those who can absorb iron normally [9]. In cases of moderate to severe anemia or malabsorption, intravenous iron may be needed, though it comes with higher costs and greater risk of side effects [10].

Oral iron therapy remains a simple, cost-effective option for mild to moderate IDA [6]. Traditionally, ferrous sulphate, fumarate, or gluconate are prescribed once daily [11]. However, gastrointestinal side effects like nausea, constipation, diarrhea, abdominal pain, and metallic taste can lower patient adherence [12–14]. Still, a pooled analysis showed that more than 60% of patients experience significant haemoglobin improvement by the second week of therapy [15].

Despite its widespread use, oral iron therapy faces two major hurdles: limited absorption and poor gastrointestinal tolerance [16]. Iron metabolism is mainly regulated by hepcidin, a hormone produced in the liver. Its levels fluctuate based on iron status, inflammation, hypoxia, and erythropoietic activity [17, 18]. Elevated hepcidin leads to degradation of ferroprotein, an iron transporter, reducing both dietary absorption and recycling of iron from old red blood cells [19, 20].

Interestingly, taking iron supplements daily, especially in the morning, keeps hepcidin levels high for nearly 24 h, lowering iron absorption from subsequent doses

[21]. Alternate-day dosing, by allowing hepcidin levels to drop between doses, has been shown to improve iron absorption [21]. A randomized trial found that taking 60 mg of elemental iron on alternate days for 28 days resulted in better absorption than daily dosing over 14 days [4]. Additionally, spacing doses out may minimize gastrointestinal side effects by reducing unabsorbed iron buildup in the gut [20, 21].

While recent studies favour alternate-day dosing, there is no clear consensus yet on whether daily or alternate-day regimens are superior for managing IDA [15]. Previous reviews often focused on specific groups, like pregnant women [13], leaving a gap in understanding the best approach for the general population. This study aims to bridge that gap by systematically reviewing and analysing evidence comparing daily versus alternate-day oral iron supplementation across diverse populations.

Methods

Protocol registration and reporting

This review evaluates the comparative effectiveness of daily and alternate-day oral iron therapies for IDA in the general population. The study followed the PRISMA guidelines for systematic reviews (supplementary Table e1). The protocol was registered with PROSPERO CRD420251014792.

Search strategy

An extensive search was carried out in several databases including PubMed, Embase, CINAHL, Scopus, and Cochrane Library. The search was updated through March 31, 2025, utilizing relevant keywords and controlled vocabulary tailored to each database (see Supplemental Table 2). Reference lists of identified articles were also reviewed for eligible studies. However, no expert consultations were conducted to find unpublished data.

Inclusion and exclusion criteria

Studies were selected based on the PICO framework and included randomized or clinical controlled trials published in English. Eligible participants were aged 10 years or older and diagnosed with IDA per WHO criteria (Hb < 13 g/dL in men, < 12 g/dL in women and adolescents ≥ 12, < 11.5 g/dL in 10–11-year-olds) [6]. We excluded children under 10 years of age because their iron requirements, absorption patterns, and anemia etiologies differ significantly from older individuals, and including this group would have introduced additional heterogeneity. Similarly, pregnant and breastfeeding women were excluded as their iron requirements are substantially different during pregnancy and lactation, and these populations are often managed according to separate clinical guidelines. Additionally, we restricted the analysis to conventional oral iron preparations such

as ferrous sulfate, ferrous fumarate, and ferrous gluconate, as newer formulations like iron-polysaccharide complexes have distinct absorption characteristics, costs, and side-effect profiles. Studies using weekly iron doses, iron-fortified foods, or lacking comparison groups were excluded. Furthermore, non-comparative designs, observational studies, cross-over trials, reviews, animal studies, case reports, and studies with incomplete data were also excluded.

The intervention group included those receiving oral iron daily in any form and dosage. Accepted formulations included iron alone or in combination with folic acid and/or vitamins. The comparator group consisted of patients receiving alternate-day oral iron.

Outcomes

The primary outcome was the difference in haemoglobin levels between the groups. Secondary outcomes included serum iron level, ferritin level, Total iron binding capacity (TIBC), mean corpuscular volume (MCV), Transferrin saturation, and incidence of adverse effects.

Data extraction

All citations retrieved from the search were uploaded into Rayyan software for screening. Duplicate entries were removed. Two reviewers independently screened studies for inclusion, with a third reviewer resolving disagreements. Daily dose and alternate-day dosing arms were compared independently for trials with multiple intervention arms.

Data collected included study title, first author, publication year, study design, country, population characteristics, sample size, intervention details, study duration, baseline and post-intervention haemoglobin and ferritin levels (with SD), and side effect frequency.

Data synthesis and analysis

A meta-analysis was carried out utilizing risk differences (RDs) for categorical data and mean differences (MDs) for continuous data. The heterogeneity was evaluated using the I^2 statistic: 0–40% (low), 30–60% (moderate), 50–90% (substantial), and 75–100% (considerable). A random-effects model was employed to ensure balanced weighting and account for differences between trials. Where applicable, subgroup and sensitivity analyses were done to examine sources of heterogeneity. Publication bias was assessed with funnel plots, with a significance level of $p < 0.05$. R Studio software, version 4.2.3 and RevMan was used for all statistical analyses. We used the “metacont” and “metabin” commands to determine the mean difference and adverse effect respectively.

Risk of bias assessment

The Cochrane Risk of Bias tool (RoB 2) was independently utilized by two reviewers to evaluate each study across domains like randomization, deviations from interventions, outcome measurement, data completeness, and selective reporting. A third reviewer settled disagreements and articles was categorized as high risk, low risk, or some concern.

Certainty of evidence

The GRADE approach was used to evaluate the certainty of the evidence for each outcome, via GRADEpro GDT software. Evidence quality was judged based on risk of bias, inconsistency, indirectness, imprecision, and publication bias.

Ethical consideration

As this review relies solely on previously published data, ethics approval was not required.

Results

Study selection and characteristics

A total 9286 articles identified in the major databases were imported in Rayyan software with 352 duplicates were removed. The remaining 8934 studies were screened based on their titles and abstracts, with 8827 articles were excluded. Full texts were retrieved for remaining 107 articles in which 11 met eligibility for inclusion in the systematic review/meta-analysis (Fig. 1).

The systematic analysis included 11 studies (a total of 1,014 participants: 507 randomized to daily iron and 507 to alternate-day iron supplementation), with the majority being from India ($n=3$), Turkey ($n=2$), Switzerland ($n=2$), Thailand, Canada, Australia, and Peru ($n=1$ each) shown in Table 1. The study designs included randomized controlled trials. Participants were mostly adults (15–62 years old), with 10 research recruiting entirely or primarily anemic people (baseline Hb: 6.5–12.8 g/dL) and two enrolling healthy cohorts; the majority of studies focused on females. The interventions employed ferrous sulfate (10 studies), ferrous fumarate, or iron sulfate, with daily dosages of 60–200 mg elemental iron and equal alternate-day dosing (except one study Kaundal et al., 2020: 60 mg daily vs. 120 mg alternate-day). Treatment length ranged from 3 weeks to 6 months (median: 8 weeks), and all trials provided pre- and post-intervention hemoglobin concentrations.

Primary outcome (haemoglobin)

An overall comparison between the endpoint values among participants receiving daily versus alternate-day interventions was done across 11 studies (involving a total of 990 participants). The pooled estimates of the endpoint mean haemoglobin values for the daily and

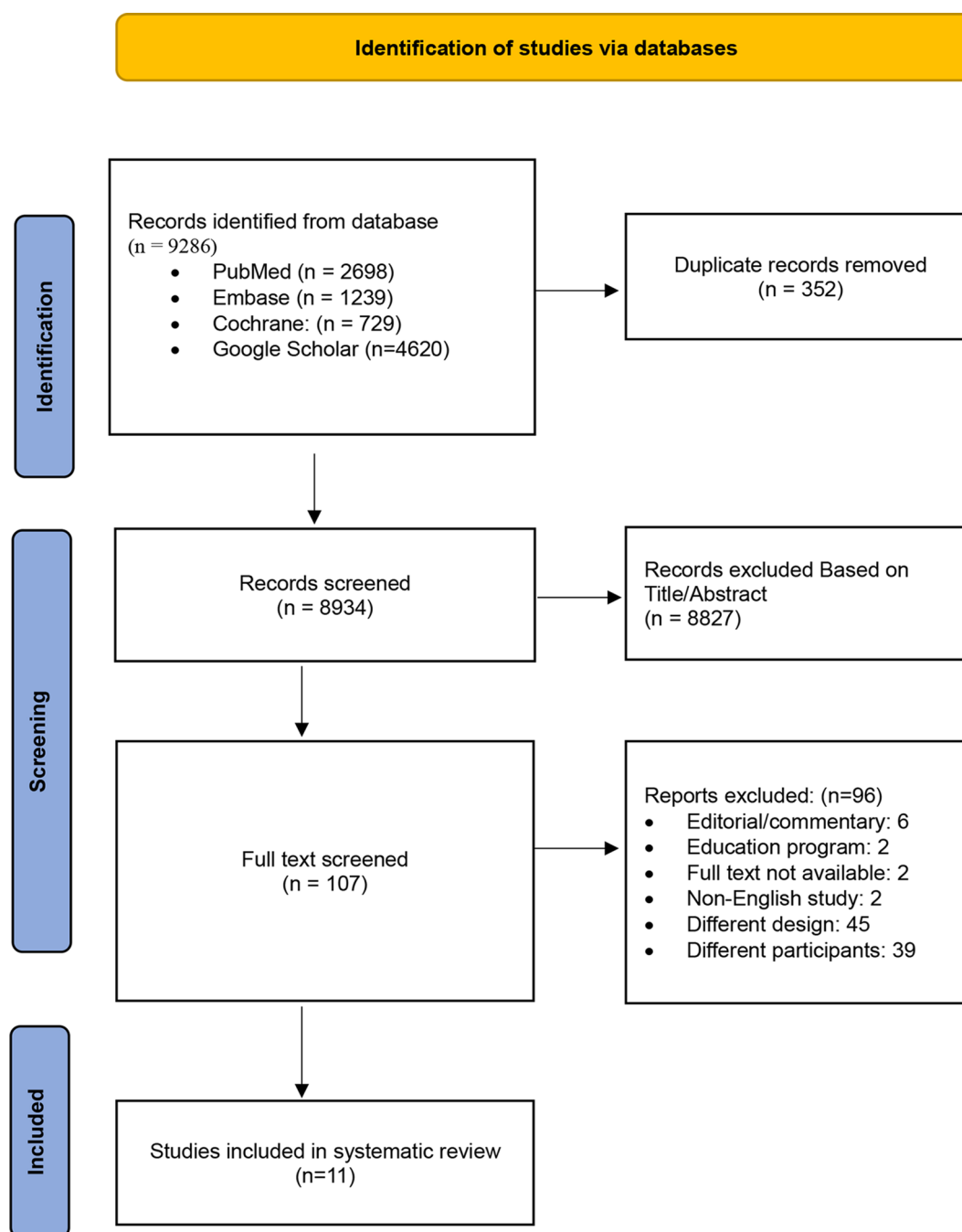


Fig. 1 Flow diagram of study selection (PRISMA)

alternate-day subgroups were 12.84 g/dL (95% CI: 12.34 to 13.34) and 12.52 g/dL (95% CI: 12.03 to 13.01) respectively. The pooled analysis showed a statistically non-significant difference favouring daily supplementation (MD: 0.28, 95% CI: -0.01 to 0.56, $p=0.06$, $z=1.91$). There was significant heterogeneity among the included studies ($I^2 = 75\%$, $\tau^2 = 0.15$, $p<0.0001$), indicating moderate variability in effect sizes (Fig. 2A). A sensitivity analysis excluding the high-risk study (McCormick et al., 2020) showed similar effect estimates for the primary outcome

with improved consistency among studies ($I^2 = 59\%$, $\text{Tau}^2 = 0.06$, $p=0.010$) shown in supplementary Fig. 1.

We compared the pooled MD with the minimal clinically important difference (MCID) for hemoglobin in iron-deficiency anemia, which is estimated to be approximately 1.0 g/dL decided by expert's consensus. The pooled MD of 0.28 g/dL does not meet this threshold, suggesting that the difference between daily and alternate-day dosing regimens, although trending towards daily dosing, is unlikely to be clinically meaningful.

Table 1 Study characteristics of included studies

Authors, Year Country	Study design	Participants				Type of iron & Days				Alternate group						
		Daily group		Alternate		Dose (f)	Hb level (before)	Hb level (after)	Dose (f)	Hb level (before)	Hb level (after)					
		Total M:F	Age (Yr)	BMI	Type							Total M:F	Age (Yr)	BMI	Type	
Jongkrai-et al., 2023 Thailand	RCT	7:25	49±15	23.7±4.7	Anae-mic	6:26	49±16	25.0±5.5	Anae-mic	ferrous fumarate Week12	200 mg	8.0±1.6	12.4±1.9	200 mg	8.1±1.9	11.9±1.1
Kaundal et al., 2020 India	RCT	5:26	37.1±12.3	-	Anae-mic	3:28	34.3±12.4	-	Anae-mic	Ferrous sulfate 6weeks	60 mg iron	8.9±1.3	11.8±1.7	120 mg iron	8.9±1.6	10.9±1.3
Kaynar et al., 2022 Turkey	RCT	F40	35.6±8.3	-	anae-mic	F43	35.8±8.4	-	anae-mic	iron sulfate 2 months	2*80 mg	9.7 ± 1.2	13.2±0.9	2*80 mg	10.1±1.5	13.3±0.7
Kron et al., 2023 Canada	RCT	4:20	(mean±SD 47.2±18.1)	(mean±SD 25.9±6.7)	anae-mic	6:21	44.4±17.1	27.0±7.0	anae-mic	Ferrous sulfate 12 weeks	300 mg (60 mg elemental iron	11 ± 1.3	13.2±1.7	300 mg (60 mg elemental iron	11 ± 1	12.6±1
McCormick et al., 2020 Australia	Parallel-group RCT	6:9	26±6	-	healthy	3:13	26±5	-	healthy	Ferrous sulfate 8weeks	105 mg of elemental iron	15.71 ± 2.46	16.48± 2.38	105 mg of elemental iron	13.42± 2.56	13.22± 1.6
Mehta et al., 2020 India	Prospective RCT	11:9	34±6.67	-	Anae-mic	12:8	33.7±7.36	-	Anae-mic	Ferrous sulfate Day 21	60 mg iron	10.05 ± 1.38	10.46±1.63	60 mg iron	9.6±1.84	11.18±1.31
Oflas et al., 2019 Turkey	Prospective RCT	F50	34.06 ± 9.593	-	Anae-mic	F50	33.2 ± 9.373	-	anae-mic	Ferrous sulfate 1 month	80 mg iron	10.48±1.38 g/dL	12.76±0.97 g/dL	80 mg iron	10.31±1.23 g/dL	12.35±1.07 g/dL
Pasupathy et al., 2023 India	RCT	98	45.33±17.31	-	anae-mic	96	49.23±15.71	-	anae-mic	Ferrous sulfate 8 weeks	60 mg elemental iron	6.68±1.89	8.01 ± 1.8	60 mg elemental iron	6.53±1.89	7.59±1.58
Siebenhentl et al., 2023 Switzerland	RCT	F75	26 (21–29)	21.7 (2.3)	anae-mic	F75	24 (21–28)	21.3 (2.1)	anae-mic	Ferrous sulfate 186 days	100 mg iron	13.3±0.8	13.4±0.8	100 mg iron	13.3±0.8	13.4±0.8

Table 1 (continued)

Authors, Year, Country	Study design	Participants		Alternate		Type of iron & Days		Daily group		Alternate group	
		Daily group		Alternate		Dose (f)	Hb level (before)	Dose (f)	Hb level (before)	Dose (f)	Hb level (before)
		Total	Age (Yr)	BMI	Type						
Stoffel et al., 2017 Switzerland	RCT	F 21	22 years (IQR 21–25)	-	anaemic	60 mg iron sulfate	Mean SD 12.8 (11.7, 14.0)	60 mg iron	Mean SD 13.2 (12.5, 14.0)	60 mg iron	Mean SD 13.6 (12.9, 14.4)
		F 101	15.3 ± 1.6	21.5 ± 3.0	healthy	60 mg iron sulfate	12.6 ± 0.94	60 mg iron	12.74 ± 0.89	60 mg iron	13.43 ± 0.89
Zavaleta et al., 2000 Peru	RCT	F 98	15.1 ± 1.6	21.3 ± 2.9	healthy	60 mg iron sulfate	13.71 ± 0.94	60 mg iron	13.71 ± 0.94	60 mg iron	13.71 ± 0.94

n = number of participants; SD = standard deviation; M = male; F = female; Yr = year; mg = milligram; f = frequency; Hb = haemoglobin; BMI = body mass index; wk = week; IQR = inter quartile range; RCT = randomized control trial;

Subgroup analysis

We conducted an exploratory subgroup analysis stratified by baseline iron status (iron-deficient vs. non-iron-deficient) and assessed the credibility of this effect using the ICEMAN criteria shown in Fig. 2B. Nine studies included participants with confirmed iron deficiency, while two studies enrolled participants without biochemical evidence of deficiency. Subgroup analyses showed no significant benefit in iron-deficient group (9 studies, *n* = 380; MD: 0.16 g/dL, 95% CI: -0.10 to 0.42; *P* = 0.22) or non-iron-deficient groups (2 studies, *n* = 230; MD: 1.68 g/dL, 95% CI: -1.23 to 4.60; *P* = 0.26) and the pooled mean difference in hemoglobin changes between daily and alternate-day dosing remained non-significant in both subgroups (*p* = 0.31). However, heterogeneity was lower in the iron-deficient group (*I*² = 61%) compared to the non-deficient group (*I*² = 94%).

Secondary outcome

An overall comparison between endpoint values of serum iron level (ug/dl) among participants receiving daily versus alternate-day intervention was conducted across 6 studies (involving 464 participants). The pooled estimate showed no significant difference between the two groups (MD: 4.64 µg/dL, 95% CI: -14.11 to 23.40, *p* = 0.657, *z* = 0.44). There was significant heterogeneity among the included studies (*I*² = 83%, *Tau*² = 441.25, *χ*² = 30.03, *df* = 5, *p* < 0.0001), reflecting considerable variability in effect sizes across different study settings and populations (supplement Fig. 2).

The pooled comparison between endpoint values of serum ferritin level (ng/ml) among participants receiving daily versus alternate-day interventions was performed conducted across 7 studies involving 664 participants. The pooled mean difference was -1.14 ng/mL (95% CI: -10.55 to 8.27), indicating no statistically significant difference between the two dosing strategies (*p* = 0.81, *Z* = 0.24). There was significant heterogeneity among the included studies (*I*² = 87%, *Tau*² = 123.52, *χ*² = 46.09, *df* = 6, *p* < 0.00001), reflecting considerable variability in effect sizes across different study settings and populations shown in Supplement Fig. 3.

An overall comparison of endpoint TIBC levels (in µg/dL) among participants receiving daily versus alternate-day interventions was performed across 3 studies (involving 176 participants). The pooled estimate showed no substantial difference across the groups (MD: -20.41 µg/dL, 95% CI: -47.27 to 6.44), *p* = 0.14, *z* = 1.49). There was a moderate heterogeneity observed among the studies (*I*² = 70%, *Tau*² = 379.04, *χ*² = 6.67, *df* = 2, *p* = 0.04), reflecting variability in effect sizes across different study settings and populations (supplement Fig. 4).

The supplement Fig. 5 shows overall comparison of transferrin saturation (%) between participants receiving

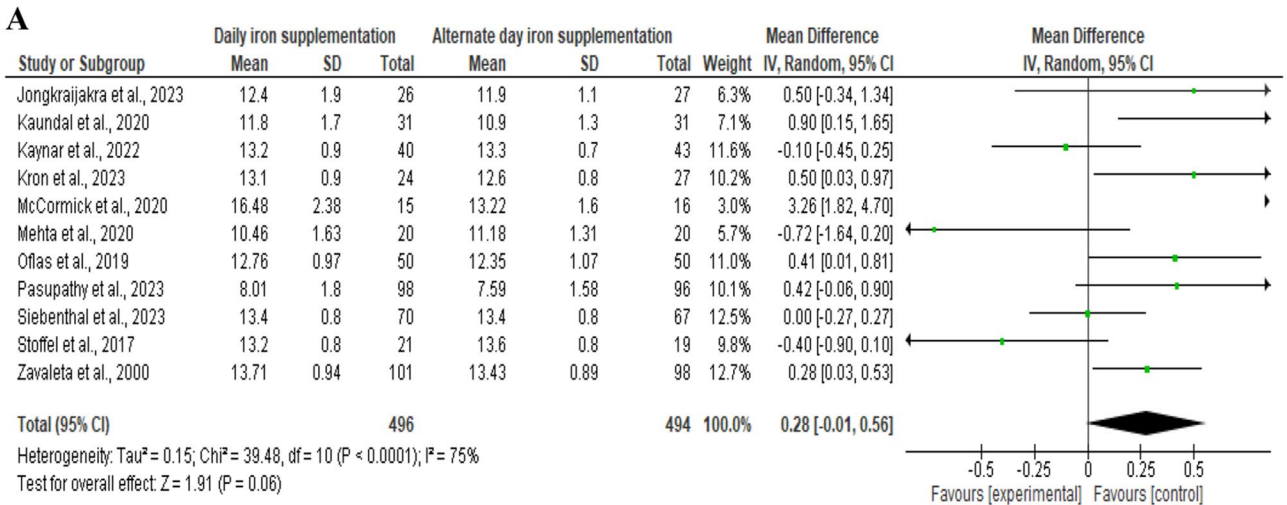


Fig. 2 A Hemoglobin outcomes in daily vs. alternate-day iron supplementation group

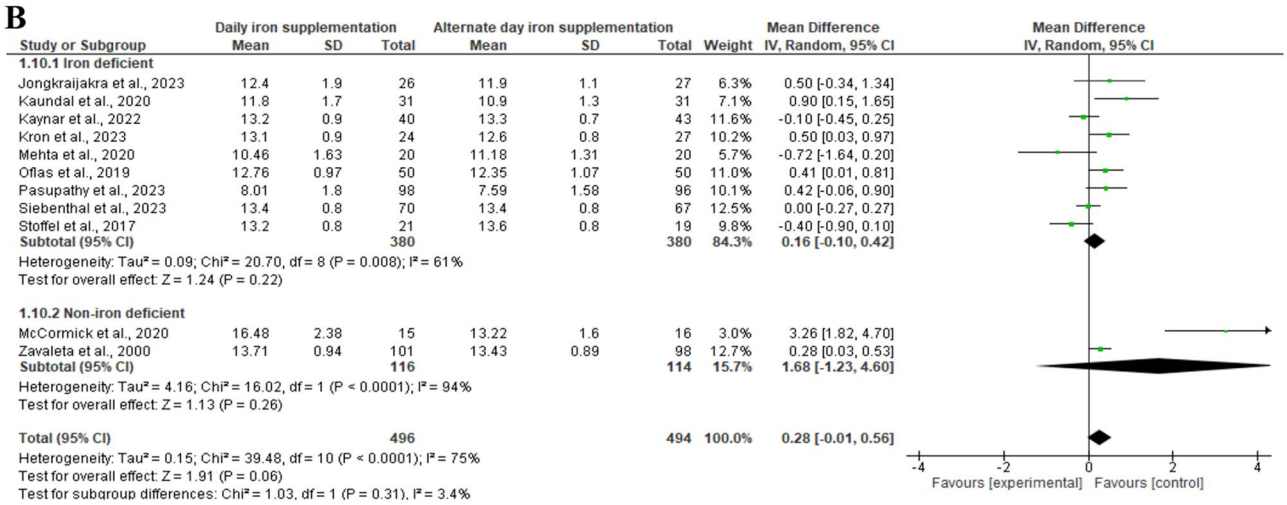


Fig. 2 B Hemoglobin outcomes in daily vs. alternate-day iron supplementation group effect using the ICEMAN criteria subgroup (iron deficient vs. non-iron deficient)

daily and alternate-day interventions was conducted across 5 studies (involving 364 participants). The pooled mean difference was -2.64% (95% CI: -13.53 to 8.26), indicating no statistically significant difference between the two dosing regimens ($p = 0.64$, $Z = 0.47$). There was a substantial heterogeneity observed among the included studies ($I^2 = 96\%$, $\tau^2 = 146.05$, $\chi^2 = 96.96$, $df = 4$, $p < 0.00001$), reflecting considerable variability in effect sizes.

An overall comparison of mean corpuscular volume (MCV) values between participants receiving daily versus alternate-day interventions was conducted across 3 studies (involving 273 participants). The pooled analysis showed no statistically significant difference across the groups (MD: 0.58 fL, 95% CI: -2.02 to 3.18 , $p = 0.66$, $Z = 0.44$). There was moderate heterogeneity ($I^2 = 75\%$,

$\tau^2 = 3.66$, $\chi^2 = 8.12$, $df = 2$, $p = 0.02$), indicating some variability in outcomes across studies (supplement Fig. 6).

Adverse effects of the iron supplementation

Figure 3 shows an overall comparison of adverse effects between daily and alternate-day iron supplementation was performed across multiple outcomes and studies. The pooled risk ratio for overall adverse effects was 1.07 (95% CI: 0.86 to 1.34), indicating no statistically significant difference between the two groups. Subgroup analyses for specific adverse effects like metallic taste, nausea, epigastric discomfort, vomiting, diarrhea, constipation, and other effects also showed no significant differences between regimens. Heterogeneity was low ($I^2 = 1.9\%$) across the pooled data, suggesting consistent results among studies. However, the test for subgroup differences was significant ($\chi^2 = 34.06$, $df = 6$, $p < 0.001$),

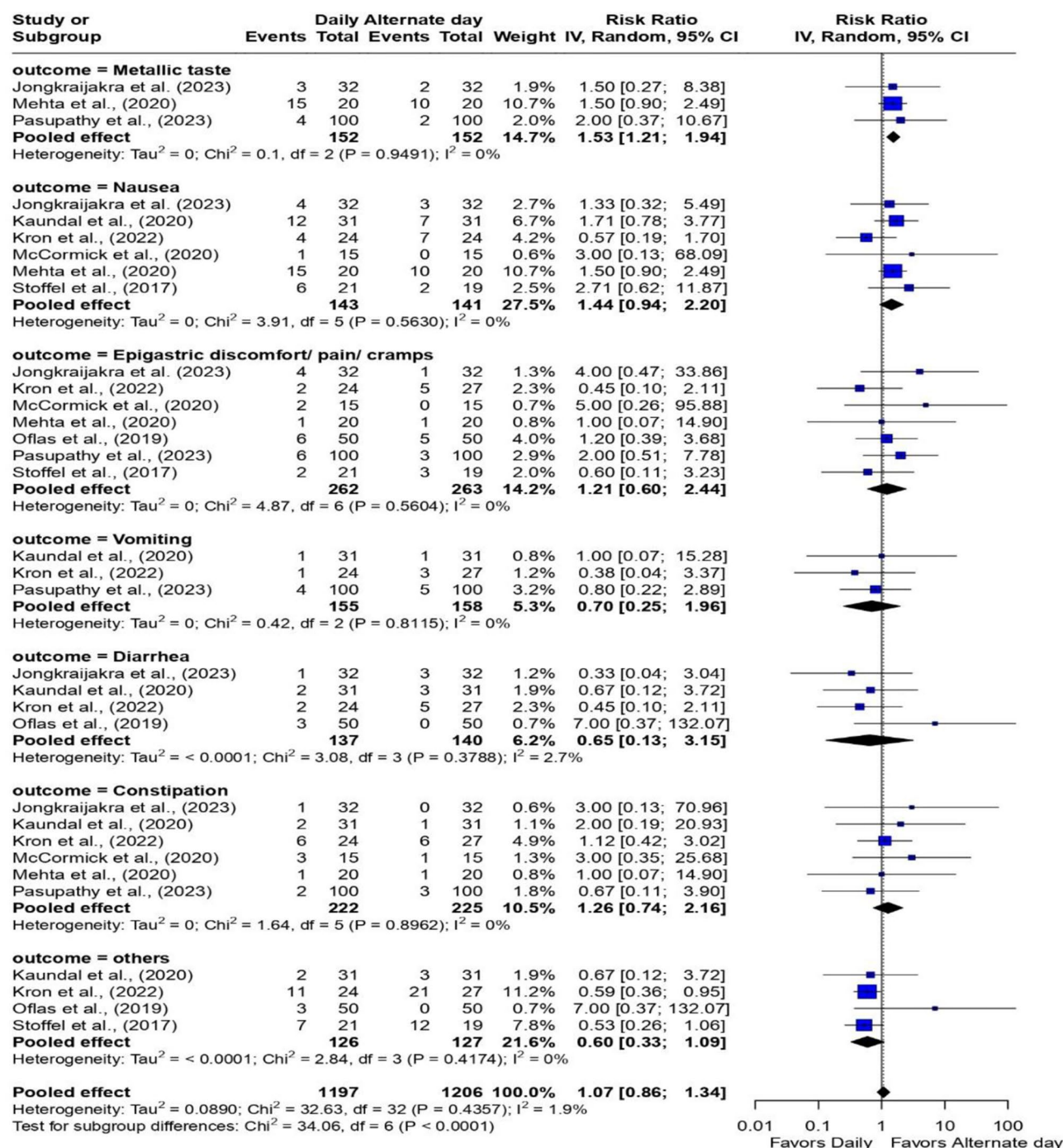


Fig. 3 Adverse outcomes in daily vs. alternate-day iron supplementation group

indicating that the type of adverse effect varied in its relative risk across regimens with outcomes metallic taste showing a higher pooled risk ratio (1.53, 95% CI: 1.21 to 1.94) favoring alternate-day dosing for reduced metallic taste, while other effects remained comparable between groups.

Publication bias

Figure 4 shows funnel plot asymmetry through the linear regression test (Egger's test) showed no evidence of significant publication bias ($t = 0.33$, $df = 9$, $p = 0.7524$). The bias estimate was 0.5476 ($SE = 1.6831$), indicating a minimal and statistically non-significant intercept. The analysis reported a residual heterogeneity variance (τ^2) of 3.1925, with inverse variance weighting applied.

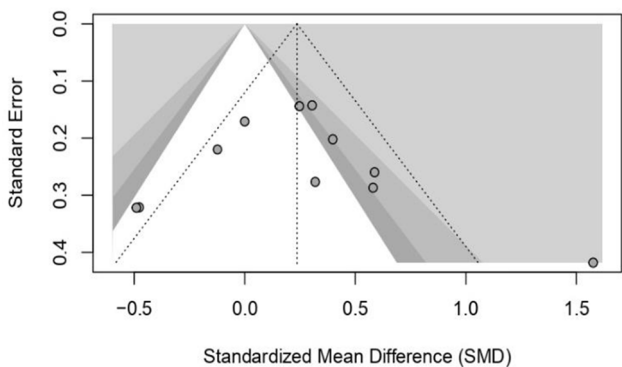


Fig. 4 Funnel plot for publication bias

Risk of bias

The risk of bias across the included randomized controlled trials was presented in Fig. 5. Most studies demonstrated a low risk of bias in domains such as outcome measurement and data completeness. However, several studies raised “some concerns” due to unclear randomization procedures or lack of blinding, and one study (Kron et al.) was rated as high risk due to methodological

limitations in allocation and intervention adherence. Despite these concerns, the overall risk of bias was considered acceptable for inclusion in the meta-analysis.

Certainty of evidence

The certainty of evidence was evaluated using the GRADE approach and presented in supplementary Table 3. Most outcomes, including hemoglobin, serum iron, ferritin, transferrin saturation, TIBC, and MCV, were rated as very low certainty due to substantial heterogeneity, imprecision, and indirectness from surrogate outcome measures. Only the outcome related to adverse effects was rated as moderate certainty, supported by consistent findings and lower heterogeneity.

Discussion

Iron deficiency anemia (IDA) is a major global health concern, particularly affecting women of reproductive age due to menstrual blood loss, dietary insufficiencies, and increased requirements during pregnancy. Oral iron supplementation remains the cornerstone of IDA treatment owing to its affordability, accessibility, and

Unique ID	D1	D2	D3	D4	D5	Overall	
Jongkrajakra et al., 2023	!	+	+	+	+	!	+ Low risk
Kaundal et al., 2020	!	!	+	+	+	!	
Kaynar et al., 2022	!	+	+	+	+	!	
Kron et al., 2023	-	+	+	+	+	-	! Some concerns
McCormick et al., 2020	!	+	+	+	+	!	- High risk
Mehta et al., 2020	!	!	+	+	!	!	
Oflas et al., 2019	!	+	+	+	!	!	
Pasupathy et al., 2023	+	+	+	+	+	+	
Siebenthal et al., 2023	+	+	+	+	+	+	
Stoffel et al., 2017	!	+	+	+	+	!	
Zavaleta et al., 2000	!	+	+	+	+	!	

Fig. 5 Risk of bias (ROB) for the included study. D1: Randomisation process; D2: Deviations from the intended interventions; D3: Missing outcome data; D4: Measurement of the outcome; D5: Selection of the reported result

established efficacy. However, the optimal dosing strategy—whether iron should be administered daily or on alternate days—has recently become a focus of clinical research and policy interest. This discussion synthesizes our findings in light of recent evidence and addresses both the strengths and inconsistencies observed in the literature.

Our meta-analysis supports comparable efficacy between daily and alternate-day iron supplementation in improving hemoglobin (Hb) concentrations, aligning with Kamath et al. [4]. However, critical contrasts emerge when contextualized with broader evidence. While trials like Pasupathy et al. [22] reported superior Hb responses and tolerability, others such as Mehta et al. [23] found no significant difference in efficacy or side effects. A contradiction potentially explained by Mehta's cohort having severe baseline anemia, which may obscure regimen-specific effects. Similarly, Kaynar et al. [24] observed no tolerability advantage with alternate-day dosing, possibly due to unstandardized adverse event reporting. These discrepancies underscore the impact of methodological heterogeneity (e.g., dosing protocols, population characteristics) on outcomes, suggesting that baseline anemia severity and inflammation status may modulate hepcidin-mediated absorption benefits. Hepcidin concentrations rise following iron ingestion and remain elevated for 24 h, inhibiting further absorption. By spacing doses, alternate-day regimens may allow for more efficient iron uptake and fewer unabsorbed iron ions in the gut, thereby minimizing adverse effects [25].

This mechanistic advantage is supported by a 2024 randomized controlled trial conducted in India, where women receiving 120 mg elemental iron on alternate days experienced a significantly greater Hb increase compared to those on 60 mg daily, with a substantially lower incidence of gastrointestinal side effects (9% vs. 45%) [22]. Similarly, a Swiss randomized trial by Stoffel et al. [9] demonstrated that alternate-day dosing resulted in similar improvements in ferritin but fewer side effects and a lower recurrence of iron deficiency over six months. These outcomes point to alternate-day regimens being more tolerable, which may translate into better adherence and improved real-world effectiveness.

However, not all studies converge on this conclusion. A 2023 randomized trial by Mehta et al. [23] involving adults with microcytic anemia found no statistically significant difference in Hb or ferritin levels between daily and alternate-day groups at eight weeks. Moreover, unlike many other studies, they observed no advantage in terms of side-effect profile with alternate-day therapy [26]. Similarly, Kaynar et al. conducted a non-randomized trial in premenopausal women and reported no clear superiority of alternate-day dosing, either in hematologic outcomes or tolerability [24].

These conflicting results may be attributed to several key variables. First, participant characteristics and baseline inflammation levels may influence iron absorption. Hepcidin levels are elevated in inflammatory states, such as metabolic syndrome or chronic infections, which may blunt the potential benefits of alternate-day dosing in certain populations [4]. Second, iron formulation and dosage matter: studies that use higher doses of iron on alternate days (e.g., 120 mg) may still lead to side effects if iron delivery exceeds absorptive capacity, despite the longer dosing interval [22, 27].

Third, methodological differences in adherence monitoring and side effect reporting could account for variable findings. In some studies, adverse effects were based on patient self-report, with no standardization, while others used structured symptom checklists. This lack of consistency may result in underestimation of gastrointestinal discomfort in the daily dosing arms [24–26]. Finally, variability in trial duration and population demographics (e.g., age, gender, dietary iron intake) may affect long-term outcomes and the perceived benefit of alternate-day therapy.

Despite these limitations, the implications of our findings are substantial. From a health promotion and implementation perspective, alternate-day dosing represents a patient-centered, biologically rational, and potentially more acceptable alternative to traditional daily therapy. This is particularly important in resource-limited settings, where tolerability influences adherence, and where health systems may lack capacity to provide intravenous iron or monitor patients closely. Alternate-day dosing may also be preferable in outpatient or community-based settings, including school health programs and maternal care services.

Nevertheless, the choice of regimen should be individualized. In cases of severe anemia or where rapid Hb correction is clinically necessary, daily dosing may still be warranted, as alternate-day regimens generally show non-inferiority rather than superiority in most hematologic indices. Furthermore, ongoing monitoring and dietary counseling should complement pharmacologic therapy to ensure sustained recovery and prevent recurrence.

Future research should address current gaps by conducting larger, multicenter trials stratified by inflammation status, anemia severity, gender, and iron formulation. Objective adherence metrics, dietary iron intake assessments, and standardized adverse event reporting must be incorporated. Studies should also explore the long-term sustainability of iron repletion, particularly in populations with high recurrence risk or chronic disease burden.

Conclusion

In conclusion, our findings support the growing evidence that alternate-day oral iron supplementation is a clinically effective and better-tolerated alternative to daily dosing for mild to moderate IDA. However, the presence of contradictory findings in the literature underscores the importance of tailoring treatment regimens to individual patient needs and contexts. With further high-quality evidence, alternate-day therapy may become a cornerstone of patient-centered anemia management strategies worldwide.

Abbreviations

IDA	Iron deficiency anemia
RoB	Risk of bias
GRADE	Grading of Recommendations Assessment Development and Evaluation
TIBC	Total iron binding capacity
MCV	Mean corpuscular volume
SRMA	systematic review and meta-analysis
RCT	Randomized control trial
MD	Mean difference
RR	Risk ratio

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40360-025-00984-2>.

Supplementary Material 1

Acknowledgements

None.

Author contributions

ADD, VP, JS: Conceptualization/ protocol registration/ Searching databases / article screening / quality assessment / data analysis / Writing – review & editing. SK, SKM, SRV: Searching databases / Data curation / data analysis / Writing – review & editing. ADD, SKM, RK: Protocol registration/ running the 'search'/ article screening / data extraction / quality assessment / Writing – original draft / Writing – review & editing. All authors have made significant contributions to the conception, design, execution, or interpretation of the study and reviewed the final version of the article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

No ethical approval is required as no human involvement. The study followed the PRISMA guidelines for systematic reviews which was registered with PROSPERO CRD420251014792.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Physiology, All India Institute of Medical Sciences, Deoghar, India

²Indian Council of Medical Research, National Institute for Implementation Research on Non Communicable Diseases, Jodhpur, India

³Department of Pharmacology, All India Institute of Medical Sciences, Deoghar, India

⁴College of Nursing, All India Institute of Medical Sciences, Deoghar, India

⁵Department of Biochemistry, All India Institute of Medical Sciences, Deoghar, Jharkhand, India

⁶Department of General Medicine, All India Institute of Medical Sciences, Deoghar, Jharkhand, India

Received: 17 June 2025 / Accepted: 4 August 2025

Published online: 21 August 2025

References

1. Uçan A, Kaya ZI, Yilmaz EÖ, Vasi İ, Özgeyik MO. Comparing therapeutic effects of alternate day versus daily oral iron in women with iron deficiency anemia: A retrospective cohort study. *Med (Baltim)*. 2023;102(30):e34421.
2. Stevens GA, Paciorek CJ, Flores-Urrutia MC, Borghi E, Namaste S, Wirth JP, Suchdev PS, Ezzati M, Rohner F, Flaxman SR, Rogers LM. National, regional, and global estimates of anaemia by severity in women and children for 2000–19: a pooled analysis of population-representative data. *Lancet Glob Health*. 2022;10(5):e627–39.
3. GBD 2021 Anaemia Collaborators. Global, regional, and National burden of anaemia and its underlying causes, 1990–2021: a systematic analysis for the global burden of disease study 2021. *Lancet Haematol*. 2024;11(4):e317–31.
4. Kamath S, Parveen RS, Hegde S, Mathias EG, Nayak V, Bolloor A. Daily versus alternate day oral iron therapy in iron deficiency anemia: a systematic review. *Naunyn Schmiedeberg Arch Pharmacol*. 2024;397(5):2701–14.
5. GBD 2019 Risk Factors Collaborators. Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the global burden of disease study 2019. *Lancet*. 2020;396(10258):1223–49.
6. World Health Organization. Haemoglobin cutoffs to define anaemia in individuals and populations: WHO guideline. Geneva: World Health Organization; 2024.
7. Johnson-Wimbley TD, Graham DY. Diagnosis and management of iron deficiency anemia in the 21st century. *Th Adv Gastroenterol*. 2011;4(3):177–84.
8. Moustarah F, Daley SF. Dietary iron. *StatPearls [Internet]*. Treasure Island (FL): StatPearls Publishing; 2024 Jan. p. 8.
9. Stoffel NU, Cercamondi CI, Brittenham G, Moretti D, Zimmermann MB. Iron absorption from oral iron supplements given on consecutive versus alternate days and as single morning doses: a randomized controlled trial. *Lancet Haematol*. 2017;4(11):e524–33.
10. Auerbach M, Adamson JW. How we diagnose and treat iron deficiency anemia. *Am J Hematol*. 2016;91(1):31–8.
11. Russo R, Campanella R, Buonuomo PS, De Angelis P, Giordano P. Oral iron therapy in pediatric patients: current landscape. *Acta Haematol*. 2020;143(4):301–10.
12. Snook J, Bhala N, Beales ILP, Cannings D, Kightley C, Logan RFA, et al. British society of gastroenterology guidelines for the management of iron deficiency anaemia in adults. *Gut*. 2021;70(11):2030–51.
13. Alleyne M, Horne MK, Miller JL. Individualized treatment for iron-deficiency anemia in adults. *Am J Med*. 2008;121(11):943–8.
14. Reveiz L, Gyte GM, Cuervo LG, Casasbuenas A. Treatments for iron-deficiency anaemia in pregnancy. *Cochrane Database Syst Rev*. 2011;(10):CD003094.
15. Cancelo-Hidalgo MJ, Castelo-Branco C, Palacios S, Haya-Palazón I, Ciria-Recasens M, Manasanch J, et al. Tolerability of different oral iron supplements: a systematic review. *Curr Med Res Opin*. 2013;29(4):291–303.
16. Okam MM, Koch TA, Tran MH. Iron supplementation, response in iron-deficiency anemia: analysis of five trials. *Am J Med*. 2016;129(10):1066–75.
17. Ganz T. Hepcidin, a key regulator of iron metabolism and mediator of anemia of inflammation. *Blood*. 2003;102(3):783–8.
18. Sangkhav V, Nemeth E. Regulation of the iron homeostatic hormone Hepcidin. *Adv Nutr*. 2017;8(1):126–36.
19. Camaschella C, Nai A, Silvestri L. Iron metabolism and iron disorders revisited in the Hepcidin era. *Haematol*. 2020;105(2):260–72.

20. Viveiros A, Panzer M, Baumgartner N, Schaefer B, Finkenstedt A, Henninger B, et al. Reduced iron export associated with Hcpidin resistance can explain the iron overload spectrum in Ferroportin disease. *Liver Int.* 2020;40(8):1941–51.
21. Sendur I, Malkan U. Comparison of efficiency and side effects of daily and alternate-day oral iron. *Egypt J Intern Med.* 2025;37(1).
22. Pasupathy R, Chandrasekaran V, Ravindran M, et al. Comparison of daily versus alternate-day oral iron in the treatment of iron deficiency anemia: a randomized controlled trial. *Int J Clin Pharmacol Ther.* 2024;62(2):101–9.
23. Mehta B, Rawat S, Shukla A. Daily vs alternate-day oral iron supplementation in iron deficiency anemia: a randomized clinical trial. *J Clin Diagn Res.* 2023;17(4):OC06–9.
24. Kaynar M, Ozdemir A, Kucuk M. Evaluation of daily vs alternate-day oral iron supplementation in premenopausal women: a non-randomized controlled study. *Turk J Med Sci.* 2023;53(1):32–8.
25. Stoffel NU, Zeder C, Brittenham GM, et al. Daily iron supplementation increases Hcpidin and decreases iron absorption: a randomized crossover trial. *Am J Clin Nutr.* 2020;111(6):1172–80.
26. Ofias N, Yildirim A, Kocoglu M. Effectiveness and Gastrointestinal side effects of different oral iron regimens in women with iron deficiency anemia. *J Obstet Gynaecol Res.* 2023;49(3):728–35.
27. Kaundal RK, Dhir M, Arora S. Comparison of efficacy and tolerability of daily vs alternate-day oral iron therapy in iron deficiency anemia: a randomized controlled trial. *Ann Hematol.* 2023;102(1):145–52.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.